Final PRACTICE Exam Solution

MA 341

Name: ______________________

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Read all of the following information before starting the exam:

- Complete each problem. Show all of your work clearly and in order and justify your answers, as partial credit will be given when appropriate and there may be NO credit given for problems without supporting work.
- **Circle or otherwise indicate your final answers.** All answers should be completely simplified, unless otherwise stated.
- **You may use your own formula sheet (one sheet - two pages long) and a *non graphical* calculator. Also see the Table of Laplace Transforms on the last page.**

1.

Consider the following initial value problem.

$$\begin{cases} -\sin(x-y)dx + \left(3y^{-1}\cos(x-y) + \sin(x-y)\right)dy & = 0 \\ y\left(0\right) & = \pi \end{cases}$$

a. Is the equation is exact? Explain!

Solution: The equation is not exact since

$$M_y = \cos\left(x-y\right) \neq \cos(x-y) - 3y^{-1}\sin(x-y) = N_x.$$

b. If you found this equation not exact, find an equivalent exact equation.

Solution: It is possible to make this equation exact using the following integration factor:

$$\mu = e^{\int \frac{N_x - M_y}{M}dy} = e^{3\int \frac{1}{y}dy} = y^3$$

Thus,

$$y^3(Mdx + Ndy) = -y^3\sin\left(x-y\right)dx + \left(3y^2\cos\left(x-y\right) + y^3\sin\left(x-y\right)\right)dy = 0.$$

c. Solve the initial problem.

Solution: Let

$$F\left(x,y\right) = -\int y^3\sin\left(x-y\right)dx = y^3\cos\left(x-y\right) + h\left(y\right)$$

Differentiate to get

$$F_y = 3y^2\cos(x-y) + y^3\sin(x-y) + h'(y) = N.$$

Hence, $h'(y) = 0$ and $h(y)$ is a constant, so we got

$$y^3 \cos(x - y) = C$$

Using IC one:

$$\pi^3 \cos(0 - \pi) = C,$$

thus $C = -\pi^3$, and finally:

$$y^3 \cos(x - y) = -\pi^3.$$

2. Find general solution of the following problems:

a. $y'' + 16y = e^{-4x} + 3\sin 4x$

Solution:

$$y = \frac{e^{-4x}}{32} + \left(-\frac{3x}{8} + c_1\right)\cos(4x) + \left(\frac{3}{64} + c_2\right)\sin(4x)$$

b. $xy' + y = y^2 x^2 \ln x$

Solution:

$$y = \left(Cx + x^2(1 - \ln x)\right)^{-1}$$

c. $y' + y \cot t = \frac{1}{\sin t}$

Solution:

$$y = \frac{t + C}{\sin(t)}$$

3.

Solve the following initial value problem:

$$\begin{cases} y'' + 4y = f(t) = \begin{cases} \sin 3t, & 0 < t < \frac{\pi}{2} \\ 0\ , & t \geq \frac{\pi}{2} \end{cases} \\ y(0) = a \\ y'(0) = b \end{cases}.$$

Solution: Rewrite $f(t)$ as a combination of step functions:

$$f(t) = \sin 3t + u_{\pi/2}(t)\,(0 - \sin 3t) = \sin 3t - u_{\pi/2}(t)\sin 3t$$

It's Laplace transform is

$$L\{f(t)\} = L\left\{\sin 3t - u_{\pi/2}(t)\sin 3t\right\} = L\{\sin 3t\} - \underbrace{L\left\{u_{\pi/2}(t)\sin 3t\right\}}_{\text{doesn't fit Laplace transform table}}$$

$$= \frac{3}{s^2+9} - L\left\{u_{\pi/2}(t)\sin\left[(3t - 3\pi/2) + 3\pi/2\right]\right\}$$

$$= \frac{3}{s^2+9} - L\left\{u_{\pi/2}(t)\left[\sin(3t-3\pi/2)\underbrace{\cos(3\pi/2)}_{=0} + \underbrace{\sin(3\pi/2)}_{=-1}\cos(3t-3\pi/2)\right]\right\}$$

$$= \frac{3}{s^2+9} - L\left\{u_{\pi/2}(t)\left[-\cos(3t-3\pi/2)\right]\right\} =$$

$$= \frac{3}{s^2+9} + \underbrace{L\left\{u_{\pi/2}(t)\cos 3(t-\pi/2)\right\}}_{\text{fits line 17}} = \frac{3}{s^2+9} + e^{-\frac{\pi}{2}s}L\{\cos 3t\} = \frac{3}{s^2+9} + e^{-\frac{\pi}{2}s}\frac{s}{s^2+9}$$

The Laplace Transform of the equation is now

$$s^2Y - sa - b + 4Y = \frac{3}{s^2+9} + e^{-\frac{\pi}{2}s}\frac{s}{s^2+9},$$

solve for Y:

$$Y = \frac{3}{(s^2+9)(s^2+4)} + e^{-\frac{\pi}{2}s}\frac{s}{(s^2+9)(s^2+4)} + \frac{sa+b}{s^2+4}$$

Using

$$\frac{1}{(s^2+9)(s^2+4)} = \frac{(s^2+9)-(s^2+4)}{5(s^2+9)(s^2+4)} = \frac{1}{5}\left(\frac{1}{s^2+4} - \frac{1}{s^2+9}\right)$$

the Inverse Laplace transform become

$$L^{-1}\left\{\frac{3}{(s^2+9)(s^2+4)}\right\} = \frac{3}{5}L^{-1}\left\{\frac{1}{s^2+4}\right\} - \frac{3}{5}L^{-1}\left\{\frac{1}{s^2+9}\right\}$$
$$= \frac{3}{5}\cdot\frac{1}{2}\sin 2t + \frac{3}{5}\cdot\frac{1}{3}\sin 3t = \frac{3}{10}\sin 2t + \frac{1}{5}\sin 3t,$$

,

$$L^{-1}\left\{e^{-\frac{\pi}{2}s}\frac{s}{(s^2+9)(s^2+4)}\right\} = \frac{1}{5}L^{-1}\left\{e^{-\frac{\pi}{2}s}\frac{s}{s^2+4}\right\} - \frac{1}{5}L^{-1}\left\{e^{-\frac{\pi}{2}s}\frac{s}{s^2+9}\right\}$$
$$= \frac{1}{5}u_{\pi/2}(t)\cos 2(t-\pi/2) - \frac{1}{5}u_{\pi/2}(t)\cos 3(t-\pi/2)$$
$$= -\frac{1}{5}u_{\pi/2}(t)\cos 2t + \frac{1}{5}u_{\pi/2}(t)\sin 3t = \frac{1}{5}u_{\pi/2}(t)(\sin 3t - \cos 2t)$$

and

$$L^{-1}\left\{\frac{sa+b}{s^2+4}\right\} = aL^{-1}\left\{\frac{s}{s^2+4}\right\} + bL^{-1}\left\{\frac{1}{s^2+4}\right\} = a\cos 2t + \frac{b}{2}\sin 2t$$

Finally,

$$y(t) = L^{-1}\{Y(s)\} = \frac{3}{10}\sin 2t + \frac{1}{5}\sin 3t + \frac{1}{5}u_{\pi/2}(t)(\sin 3t - \cos 2t) + C_1\cos 2t + \frac{C_2}{2}\sin 2t$$
$$= \left(\frac{3}{10} + \frac{b}{2}\right)\sin 2t + \frac{1}{5}\sin 3t + a\cos 2t + \frac{1}{5}u_{\pi/2}(t)(\sin 3t - \cos 2t)$$

4. Solve the following system of equations:

$$\begin{cases} y' - 4y - v &= t \\ v' - 2y - 3v &= -t \end{cases}$$

Solution: Rewrite the system as $\mathbf{x}' = A\mathbf{x} + \mathbf{f}$:

$$\begin{pmatrix} y' \\ v' \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} y \\ v \end{pmatrix} + \begin{pmatrix} t \\ -t \end{pmatrix}$$

Find eigenvalues of the Matrix

$$\det(A - rI) = det \begin{pmatrix} 4-r & 1 \\ 2 & 3-r \end{pmatrix} = (4-r)(3-r) - 2 = (r-2)(r-5)$$

Find eigenvector associated with $r_1 = 2$:

$$\begin{pmatrix} 4-2 & 1 \\ 2 & 3-2 \end{pmatrix} v_1 = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} v_1 = 0 \Rightarrow v_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Find eigenvector associated with $r_2 = 5$:

$$\begin{pmatrix} 4-5 & 1 \\ 2 & 3-5 \end{pmatrix} v_2 = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} v_v = 0 \Rightarrow v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Thus we got

$$y_h = c_1 v_1 e^{r_1 t} + c_2 v_2 e^{r_2 t} = c_1 \begin{bmatrix} 1 \\ -2 \end{bmatrix} e^{2t} + c_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{5t} = \underbrace{\begin{bmatrix} e^{2t} & e^{5t} \\ -2e^{2t} & e^{5t} \end{bmatrix}}_{=X} \underbrace{\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}}_{\mathbf{c}}$$

To find y_p we employ variation of parameters for systems. We first need to solve $X\mathbf{u}' = \mathbf{f}$. Using Cramer's rule one get:

$$u_1' = \frac{\det \begin{bmatrix} t & e^{5t} \\ -t & e^{5t} \end{bmatrix}}{\det X} = \frac{2te^{5t}}{3e^{7t}} = \frac{2}{3} t e^{-2t}, \quad u_2' = \frac{\det \begin{bmatrix} e^{2t} & t \\ -2e^{2t} & -t \end{bmatrix}}{\det X} = \frac{te^{2t}}{3e^{7t}} = \frac{1}{3} t e^{-5t}.$$

Thus

$$
\begin{aligned}
y_p = X\mathbf{u} = X\int \mathbf{u}'(\mathbf{s})ds \\
&= \frac{1}{3}X\int \begin{bmatrix} 2se^{-2s} \\ \\ se^{-5s} \end{bmatrix} ds = -\frac{1}{3}\begin{bmatrix} e^{2t} & e^{5t} \\ \\ -2e^{2t} & e^{5t} \end{bmatrix}\begin{bmatrix} 2\frac{e^{-2t}(2t+1)}{2^2} \\ \\ \frac{e^{-5t}(5t+1)}{5^2} \end{bmatrix} \\
&= -\frac{1}{3}\begin{bmatrix} \frac{2t+1}{2} + \frac{5t+1}{25} \\ \\ -(2t+1) + \frac{5t+1}{25} \end{bmatrix} \\
&= -\frac{1}{150}\begin{bmatrix} 60t+27 \\ \\ -90t-48 \end{bmatrix} = -\frac{1}{50}\begin{bmatrix} 20t+9 \\ \\ -30t-16 \end{bmatrix}
\end{aligned}
$$

Good luck!

Table of Laplace Transforms

N	f(t) $= \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1	1	$\frac{1}{s},\ s > 0$
2	e^{at}	$\frac{1}{s-a},\ s > a$
3	$t^n, n \in \mathbb{N}$	$\frac{n!}{s^{n+1}},\ s > 0$
4	$\sin at$	$\frac{a}{s^2+a^2},\ s > 0$
5	$\cos at$	$\frac{s}{s^2+a^2},\ s > 0$
6	$\sinh at$	$\frac{a}{s^2-a^2},\ s > \|a\|$
7	$\cosh at$	$\frac{s}{s^2-a^2},\ s > \|a\|$
8	$e^{at}\sin bt$	$\frac{b}{(s-a)^2+b^2},\ s > a$
9	$e^{at}\cos bt$	$\frac{s-a}{(s-a)^2+b^2},\ s > a$
10	$t^n e^{at}, n \in \mathbb{N}$	$\frac{n!}{(s-a)^{n+1}},\ s > 0$
11	$e^{at}f(t)$	$F(s-a)$
12	$f^{(n)}(t)$	$s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$
13	$(-t)^n f(t)$	$F^{(n)}(s)$
14	$t^n f(t)$	$(-1)^n F^{(n)}(s)$
15	$f(at)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
16	$u_a(t)$	$\frac{e^{-as}}{s}$
17	$u_a(t)f(t-a)$	$e^{-as}F(s)$
18	$\delta(t-a)$	e^{-as}